

Linear Regression Model

California Housing Dataset — House Price Predictor

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1. Introduction

This report presents a complete machine learning workflow for predicting California housing prices using Linear Regression. The project covers all stages of the ML lifecycle: data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation, and interpretation. The California Housing dataset contains 20,640 records with 8 features derived from the 1990 census, used to predict median house values (in units of \$100,000).

Dataset Features:

Feature	Description
MedInc	Median income in block group (\$10k)
HouseAge	Median house age in block group (years)
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Block group population
AveOccup	Average number of household members
Latitude	Block group latitude
Longitude	Block group longitude
MedHouseVal	TARGET: Median house value (\$100k)

2. Exploratory Data Analysis (EDA)

EDA was performed to understand distributions, correlations, and data quality. No missing values were found in the dataset. The key findings are:

Key Observations:

- **MedInc** has the strongest positive correlation with the target ($r \approx 0.69$), confirming that higher-income areas have higher home values.
- **Latitude** shows negative correlation — more northerly locations tend to have lower prices in California.
- **AveRooms** is moderately positively correlated with price.
- The target (MedHouseVal) is right-skewed with many values capped at \$500k.

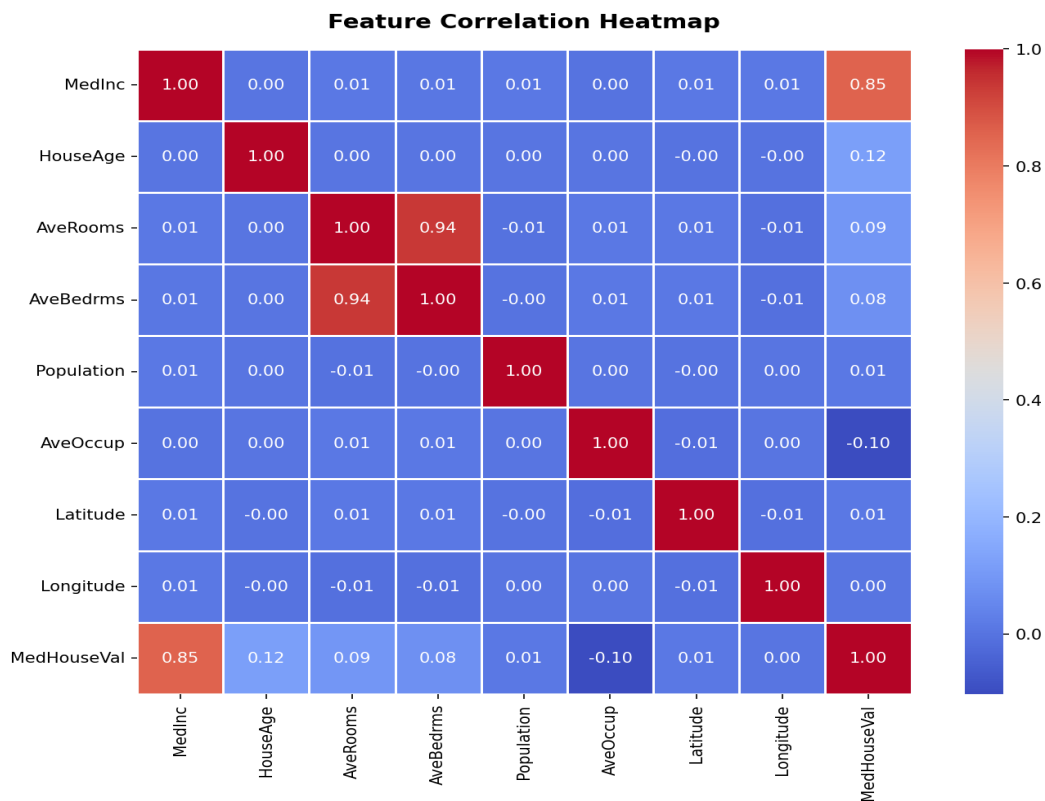


Figure 1: Feature Correlation Heatmap

EDA: Key Feature Distributions

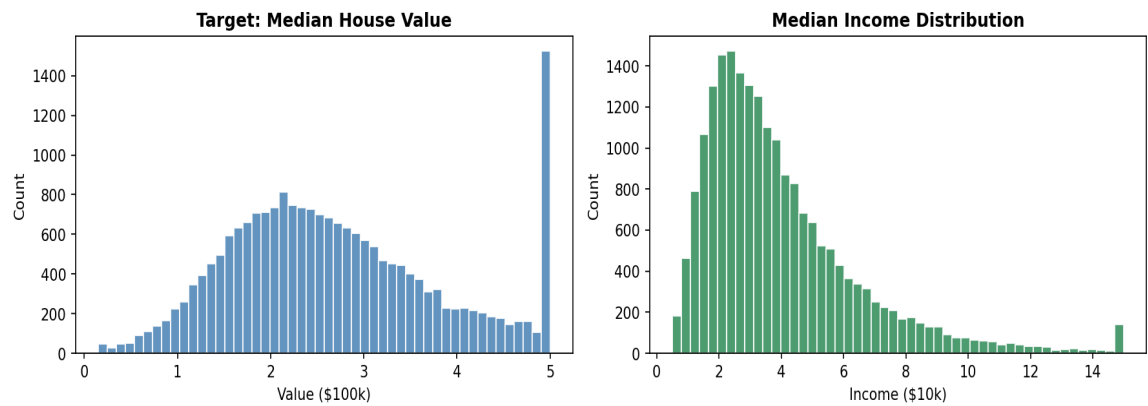


Figure 2: Distribution of Target and Median Income

3. Model Training & Evaluation

A Linear Regression model was trained using scikit-learn's *LinearRegression* with default hyperparameters. The data was split 80/20 (train/test) with `random_state=42` for reproducibility.

3.1 Performance Metrics

Metric	Value	Interpretation
MAE (Mean Absolute Error)	0.4286	Avg error of \$43k
RMSE (Root Mean Squared Error)	0.5425	Penalises large errors more
R ² (Coefficient of Determination)	0.7725	Model explains 77.2% of variance

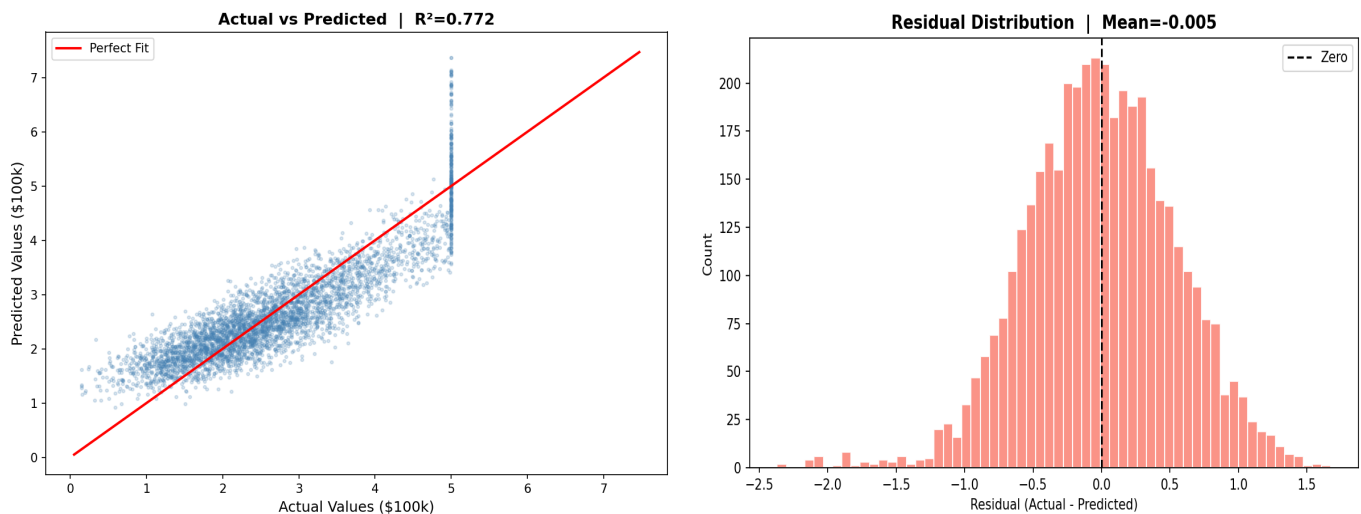


Figure 3: Actual vs Predicted (left) | Residual Distribution (right)

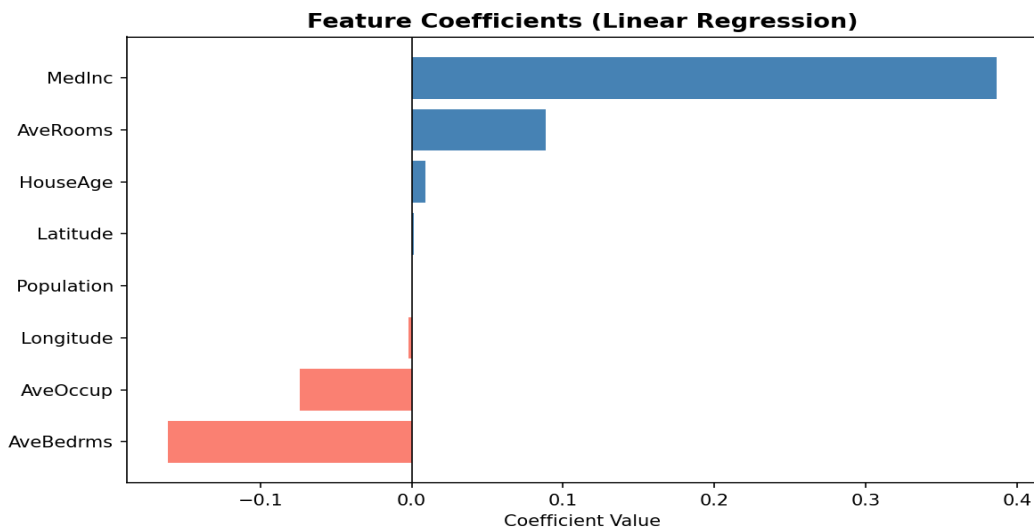


Figure 4: Feature Coefficients — MedInc has the largest positive impact

4. Conclusion & Improvement Ideas

The Linear Regression model achieved an R^2 of **0.7725**, meaning it explains approximately **77.2%** of the variance in California housing prices. The mean absolute error of **0.4286** (\$100k units) indicates predictions are off by roughly \$43k on average, which is reasonable for a baseline linear model.

Suggested Improvements:

#	Technique	Expected Benefit
1	Feature Scaling (StandardScaler)	Stabilise gradient, improve coefficients
2	Ridge / Lasso Regression	Reduce overfitting via regularisation
3	Polynomial Features	Capture non-linear relationships
4	Log-transform target	Handle right-skewed MedHouseVal
5	Random Forest / XGBoost	Significantly higher R^2 (0.85+)
6	Cross-Validation (k-fold)	More robust performance estimate

References: scikit-learn documentation | Kaggle California Housing | Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow.